Code No: R203105A **R20** SET - 1

III B. Tech I Semester Regular Examinations, Dec/Jan – 2022-23
ARTIFICIAL INTELLIGENCE
(Common to CSE, IT)

Time: 3 hours Max. Marks: 70

Answer any **FIVE** Questions **ONE** Question from **Each unit**
All Questions Carry Equal Marks
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UNIT-I

1. a) Discuss categorization of intelligent systems. [7M]
   b) Explain in detail the applications of Artificial Intelligence. [7M]

(OR)

2. a) Discuss the characteristics of AI problem. [7M]
   b) How can a Tic Tac Toe game application is created using AI? [7M]

UNIT-II

3. a) Solve the water-jug problem by writing the production rules. [7M]
   b) Explain various blind search strategies. [7M]

(OR)

4. a) Explain about A* algorithm in detail. [7M]
   b) Define constraint satisfaction problem (CSP). How CSP is formulated as a search problem? Explain with an example. [7M]

UNIT-III

5. a) Give the rules of inference in propositional logic and explain natural deduction using an example. [7M]
   b) Solve the given propositional calculus expressions are equivalent or not (P->Q->R) AND (P->Q ^ Q->R) [7M]

(OR)

6. a) Define Axiomatic system. State the axioms and the rules used in the Axiomatic system. [7M]
   b) What is predicate logic? Explain the predicate logic representation with reference to suitable example. [7M]

UNIT-IV

7. a) Discuss about Knowledge Representation using Semantic Network. [7M]
   b) Represent the following knowledge using a semantic network [7M]
   Anil is a writer
   Anil lives in Bombay
   Swathi is a teacher
   Swathi lives in Bangalore.
   Anil sent a copy of his book to Swathi
   Swathi sent his thanks to Anil.

(OR)

8. a) Explain about Extended semantic networks for KR. [7M]
   b) Explain about the rules for conceptual dependencies. [7M]

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UNIT-V

9.	a)	With the help of a neat diagram, explain the Expert System Architecture	[7M]
	b)	What is Inference Engine? Describe Backward and Forward chaining mechanism used by an inference engine?	[7M]
		(OR)	
10.	a)	How is an expert system different from a traditional program?	[7M]
	b)	List the characteristics of expert systems. Classify various Expert System shells and tools.	[7M]

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Code No: R203105A **R20** SET - 2

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Time: 3 hours Max. Marks: 70

Answer any **FIVE** Questions **ONE** Question from **Each unit**
All Questions Carry Equal Marks
*****

UNIT-I

1. a) What is AI? Explain any four approaches to AI. [7M]
   b) Explain the techniques of A.I. Also describe the characteristics of Artificial Intelligence. [7M]

(OR)

2. a) How can a Tic Tac Toe game application is created using AI? [7M]
   b) List and explain the applications of Artificial Intelligence. [7M]

UNIT-II

3. a) Draw a state space representation of Towers of Hanoi problem. [7M]
   b) Provide a state space for the game of chess. [7M]

(OR)

4. a) Explain about A* algorithm in detail. [7M]
   b) Illustrate the heuristic Hill climbing algorithm with an example. [7M]

UNIT-III

5. a) Show by using truth table the expressions are logical equivalent [7M]
   $[(A \rightarrow B) \rightarrow C, A \rightarrow (B \rightarrow C)]$ and $[(A \wedge \sim B) \rightarrow C, \sim (A \wedge \sim B \wedge \sim C)]$.
   b) Compare and contrast the two variants of logic-predicate and propositional. [7M]

(OR)

6. a) Illustrate semantic tableau system in propositional logic using the rules? [7M]
   b) What is predicate logic? Explain the predicate logic representation with reference to suitable example. [7M]

UNIT-IV

7. a) Explain Inheritance in Semantic Net. [7M]
   b) Illustrate the Backward Reasoning Inference Method by using some Example. [7M]

(OR)

8. a) Explain different conceptual Primitive Actions. [7M]
   b) What is meant by Script? Write a script for Going to Theater. [7M]

UNIT-V

9. a) What are the prominent features of an expert system and describe their features in detail. [7M]
   b) Explain Architecture of an Expert system. Give its three application areas. [7M]

(OR)

10. a) Describe different type of knowledge required to build an expert system. [7M]
    b) Give an overview of various types of expert system tools and criteria for selecting the right kind of tool. [7M]

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Code No: R203105A **R20** SET - 3

III B. Tech I Semester Regular Examinations, Dec/Jan – 2022-23
ARTIFICIAL INTELLIGENCE
(Common to CSE, IT)

Time: 3 hours Max. Marks: 70

Answer any **FIVE** Questions **ONE** Question from **Each unit**
All Questions Carry Equal Marks
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UNIT-I

1.	a)	Briefly Explain the history of Artificial Intelligence.	[7M]
	b)	What is AI? Explain any four approaches to AI.	[7M]
		(OR)	
2.	a)	How can a Tic Tac Toe game application is created using AI?	[7M]
	b)	What are current trends in A.I	[7M]

UNIT-II

3.	a)	Solve the water-jug problem by writing the production rules.	[7M]
	b)	Describe different control strategies used in problem solving.	[7M]
		(OR)	
4.	a)	Explain about A* algorithm in detail.	[7M]
	b)	Define constraint satisfaction problem (CSP). How CSP is formulated as a search problem? Explain with an example.	[7M]

UNIT-III

5.	a)	Explain the forward-chaining algorithm for propositional logic.	[7M]
	b)	Apply resolution refutation in proportional logic for checking the equivalence of expressions.	[7M]
		(OR)	
6.	a)	Define Axiomatic system. State the axioms and the rules used in the Axiomatic system.	[7M]
	b)	Prove the following theorem using deductive inference rules From $A \rightarrow B \wedge C$, A infer C, from $A \wedge B$, $A \rightarrow C$ infer C.	[7M]

UNIT-IV

7.	a)	Explain Inheritance Rules in Prolog.	[7M]
	b)	Illustrate the Forward Reasoning Inference Method by using some Example.	[7M]
		(OR)	
8.	a)	Explain the rules for Conceptualization Blocks in CD.	[7M]
	b)	What is meant by Script? Write a script for Restaurant Problem.	[7M]

UNIT-V

9.	a)	Explain the phases in building expert system.	[7M]
	b)	Discuss the expert system in domain of medicine using suitable case study? Explain its architecture describing its components.	[7M]
		(OR)	
10.	a)	Give two examples of non-monotonic system. Consider some monotonic and non monatomic applications and show how you can solve them using truth monotonic system.	[7M]
	b)	Build a rule based expert system for criminal identification.	[7M]

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Code No: R203105A **R20** SET - 4

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Time: 3 hours Max. Marks: 70

Answer any **FIVE** Questions **ONE** Question from **Each unit**
All Questions Carry Equal Marks
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UNIT-I

1. a) Define Artificial Intelligence. Explain the techniques of A.I [7M]
   b) Discuss categorization of intelligent systems. [7M]

(OR)

2. a) How can a Tic Tac Toe game application is created using AI? [7M]
   b) List and explain the applications of Artificial Intelligence. [7M]

UNIT-II

3. a) Draw a state space representation of Towers of Hanoi problem. [7M]
   b) Describe the factors determining the choice of direction of a particular problem. [7M]

(OR)

4. a) Explain the Heuristic Search Techniques. [7M]
   b) Explain about A* algorithm in detail. [7M]

UNIT-III

5. a) Give the rules of inference in propositional logic and explain natural deduction using an example. [7M]
   b) Explain the syntax and semantics of propositional logic. [7M]

(OR)

6. a) Show by using truth table the expressions are logical equivalent [(A → B) →C, A→ (B → C)] and [(A ∧ ~ B) →C, ~ (A ∧ ~ B ∧ ~ C)]. [7M]
   b) Illustrate semantic tableau system in propositional logic using the rules? [7M]

UNIT-IV

7. a) Give an overview of various knowledge representation approaches. [7M]
   b) Represent the following knowledge using a semantic network [7M]
      Tom is a Cat
      Tom caught a bird
      Tom is owned by John
      Tom is ginger in colour
      Cats like cream
      Cat is on the mat
      A Cat is a Mammal
      A Bird is an animal
      All mammals are animals
      Mammals have fur

(OR)

8. a) Explain Conceptual Parsing briefly. [7M]
   b) What is meant by Script? Write a script for Going to Theater. [7M]

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UNIT-V

9.	a)	Briefly explain the architecture of expert systems.	[7M]
	b)	Explain the Applications of the Expert systems.	[7M]
		(OR)	
10.	a)	Explain about MYCIN Expert system in detail.	[7M]
	b)	Explain the Issues in black board systems for problem solving.	[7M]